

SwinLLIE: Swin Transformer for Low-Light Image Enhancement with Multi-Scale Processing

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Abstract—Low-light image enhancement is essential for downstream computer vision tasks including autonomous driving, surveillance, and semantic segmentation. Conventional methods often struggle with non-uniform illumination, leading to over-exposure, color distortion, and texture loss. To address these limitations, we propose SwinLLIE (Swin Transformer for Low-Light Image Enhancement), a novel architecture that combines the efficiency of Swin Transformers with a U-Net encoder-decoder structure for multi-scale feature processing. Our method leverages window-based self-attention for computational efficiency while employing a hybrid loss function incorporating L1 reconstruction, VGG perceptual, color consistency, edge preservation, and exposure control losses. Experimental results on the LOL dataset demonstrate that SwinLLIE achieves a PSNR of 18.66 dB and SSIM of 0.84 with a compact model size of approximately 4.7M parameters. The proposed architecture effectively balances brightness enhancement with structural preservation, producing visually coherent and perceptually enhanced results.

Index Terms—Low-Light Enhancement, Swin Transformer, Image Restoration, Deep Learning, Multi-Scale Processing, Window Attention

I. INTRODUCTION

Low-light image enhancement plays a crucial role in downstream computer vision applications such as autonomous driving, semantic segmentation, and security surveillance. Conventional enhancement methods often overlook non-uniform illumination handling, which leads to overexposure, detail loss, and texture degradation within the brightness distribution. To address these limitations, SwinLLIE (Swin Transformer for Low-Light Image Enhancement) is proposed with a novel hybrid CNN-Transformer architecture that leverages multi-scale spatial feature processing to achieve improved perceptual quality and structural integrity in enhanced low-light images.

Unlike traditional methods that apply uniform brightness adjustments, SwinLLIE introduces a U-Net-based generator [14] with Swin Transformer blocks for multi-scale feature extraction. The Swin Transformer architecture, originally designed for image classification, has demonstrated remarkable success in image restoration tasks due to its ability to capture both local and global dependencies through shifted window-based self-attention mechanisms [1], [2].

The key contributions of this work are:

- A hybrid CNN-Swin Transformer architecture adapted for low-light enhancement with an encoder-decoder structure featuring skip connections

- An efficient window-based attention mechanism that reduces computational complexity from $O(N^2)$ to $O(N \times w^2)$ where w is the window size
- A systematic study of Swin Transformer adaptation for low-light enhancement, providing insights for future architectures
- Analysis of hybrid loss functions for balancing reconstruction and perceptual quality in illumination enhancement tasks

II. LITERATURE REVIEW

Recent deep learning-based methods have shown significant advancements in the low-light image enhancement space. Traditional enhancement methods, such as histogram equalization and Retinex-based methods, tend to struggle with non-uniform lighting conditions, which often result in unnatural detail and structural incoherency [3]. Deep learning models such as LLNet, SIDNet, and Retinex-Net enhance brightness and suppress noise, but their reliance on spatial-domain processing can produce over-smoothing and details loss [4].

GAN-based methods such as EnlightenGAN and Zero-DCE use perceptual enhancement but still struggle with preserving fine textures and intricate details in challenging lighting conditions [5], [6].

Transformer-based approaches have recently emerged as powerful alternatives. The Swin Transformer [1] introduced hierarchical feature maps and shifted window-based attention, enabling efficient computation for high-resolution images. SwinIR [2] extended this to image restoration tasks, demonstrating state-of-the-art performance in super-resolution and denoising. However, its direct application to low-light enhancement requires architectural adaptations to handle the unique challenges of illumination variation.

III. PROPOSED SWINLLIE ENHANCEMENT METHOD

A. Architecture Overview

SwinLLIE enhances low-light images by utilizing multi-scale spatial feature extraction through a U-Net encoder-decoder architecture [14] powered by Swin Transformer blocks. The architecture consists of three main stages: shallow feature extraction, deep feature extraction through encoder-decoder processing, and image reconstruction.

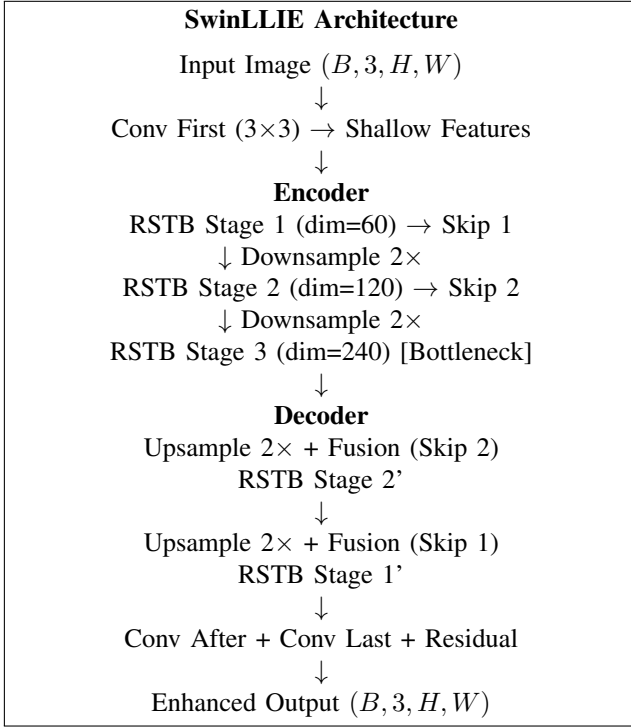


Fig. 1. High-level architecture of SwinLLIE. The generator follows an encoder-decoder structure with RSTB (Residual Swin Transformer Block) stages at multiple scales with skip connections for detail preservation.

B. Window-based Self-Attention

The core of SwinLLIE is the window-based multi-head self-attention (W-MSA) mechanism. Instead of computing attention across the entire image, which has quadratic complexity, we partition the feature map into non-overlapping windows and compute attention within each window:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}} + B\right)V \quad (1)$$

where Q, K, V are the query, key, and value matrices, d is the dimension per head, and B is the relative position bias [1]. The attention mechanism builds upon the standard scaled dot-product attention proposed by Vaswani et al. [8]. For an image of size $H \times W$ with window size $w = 8$, window attention reduces complexity from $O((HW)^2)$ to $O(HW \times w^2)$ [1].

To enable cross-window connections, shifted window attention (SW-MSA) is alternated with regular window attention, allowing information flow between adjacent windows [1].

C. Residual Swin Transformer Block (RSTB)

D. Residual Swin Transformer Block (RSTB)

Each RSTB, adopted from SwinIR [2], contains a stack of Swin Transformer blocks followed by a convolutional layer with a residual connection:

$$\text{RSTB}(X) = \text{Conv}(\text{BasicLayer}(X)) + X \quad (2)$$

The BasicLayer consists of alternating W-MSA and SW-MSA blocks:

$$\begin{aligned} X' &= \text{W-MSA}(\text{LN}(X)) + X \\ X'' &= \text{MLP}(\text{LN}(X')) + X' \\ Y' &= \text{SW-MSA}(\text{LN}(X'')) + X'' \\ Y &= \text{MLP}(\text{LN}(Y')) + Y' \end{aligned} \quad (3)$$

E. Feature Fusion for Skip Connections

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To effectively combine encoder and decoder features, we employ a learnable fusion module regarding the skip connections concept [14]:

$$F_{\text{fused}} = F_{\text{enc}} + F_{\text{dec}} + \text{Conv}([F_{\text{enc}}, F_{\text{dec}}]) \quad (4)$$

where $[\cdot, \cdot]$ denotes concatenation along the channel dimension. This three-way combination (addition + learned fusion) enables effective multi-scale feature integration while preserving fine details from the encoder.

IV. LOSS FUNCTIONS

We employ a hybrid loss function that balances reconstruction accuracy with perceptual quality:

A. L1 Reconstruction Loss

The primary reconstruction loss [16] ensures pixel-level accuracy:

$$\mathcal{L}_{L1} = \mathbb{E}[\|I_{\text{pred}} - I_{\text{gt}}\|_1] \quad (5)$$

B. VGG Perceptual Loss

To preserve high-level features and textures, we compute the L2 distance in VGG19 feature space [9]:

$$\mathcal{L}_{VGG} = \sum_i \|\phi_i(I_{\text{pred}}) - \phi_i(I_{\text{gt}})\|_2^2 \quad (6)$$

where ϕ_i represents features from pretrained VGG19.

C. Color Consistency Loss

To prevent color shifts, we adopt the color consistency loss from [6] and compute per-pixel cosine similarity:

$$\mathcal{L}_{\text{color}} = 1 - \frac{1}{HW} \sum_{h=1}^H \sum_{w=1}^W \frac{\mathbf{I}_{\text{pred}}(h, w) \cdot \mathbf{I}_{\text{gt}}(h, w)}{\|\mathbf{I}_{\text{pred}}(h, w)\|_2 \|\mathbf{I}_{\text{gt}}(h, w)\|_2 + \epsilon} \quad (7)$$

where $\mathbf{I}(h, w) \in \mathbb{R}^3$ denotes the RGB vector at spatial location (h, w) and $\epsilon = 10^{-8}$ prevents division by zero.

D. Edge Preservation Loss

Using Sobel operators S_x and S_y for edge detection [16]:

$$\mathcal{L}_{\text{edge}} = \|E(I_{\text{pred}}) - E(I_{\text{gt}})\|_1 \quad (8)$$

where $E(I) = \sqrt{(S_x * I)^2 + (S_y * I)^2}$.

E. Exposure Control Loss

To prevent overexposure while preserving bright regions, we utilize the exposure control loss [6]:

$$\mathcal{L}_{exp} = \|\text{ReLU}(I_{\text{pred}} - \tau_h)\|_2^2 + \lambda_b \|M_b \odot (I_{\text{pred}} - I_{\text{gt}})\|_1 \quad (9)$$

where $\tau_h = 0.95$ is the highlight threshold, $M_b = \mathbf{1}[I_{\text{gt}} > 0.8]$ is a binary mask identifying bright regions in the ground truth, and $\odot$ denotes element-wise multiplication.

F. Total Loss

The combined objective function is:

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{L1} + \lambda_v \mathcal{L}_{VGG} + \lambda_c \mathcal{L}_{color} + \lambda_e \mathcal{L}_{edge} + \lambda_x \mathcal{L}_{exp} \quad (10)$$

with default weights $\lambda_1 = 1.0$, $\lambda_v = 0.1$, $\lambda_c = 0.5$, $\lambda_e = 1.0$, $\lambda_x = 1.0$.

V. EXPERIMENTAL EVALUATION

A. Implementation Details

SwinLLIE was implemented using PyTorch. The model was trained for up to 100 epochs using the AdamW optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$), with the best performing model selected based on validation scores. Training was performed on random patches of size 96×96 with a batch size of 8. The detailed training hyperparameters are listed in Table I.

TABLE I
TRAINING HYPERPARAMETERS

Hyperparameter	Value
Optimizer	AdamW
Learning Rate	2×10^{-4}
Weight Decay	1×10^{-4}
Learning Rate Scheduler	Cosine Annealing
Warmup Epochs	5
Minimum Learning Rate	1×10^{-6}
Patch Size	96×96
Batch Size	8
Gradient Clipping	1.0
Loss Weights (λ_{l1} , λ_{vgg} , λ_{color})	1.0, 0.1, 0.5
Loss Weights (λ_{edge} , $\lambda_{exposure}$)	1.0, 1.0

B. Model Configuration

TABLE II
SWINLLIE ARCHITECTURE CONFIGURATION

Parameter	Value
Embedding dimension	60
Depths (encoder stages)	[4, 4, 4]
Number of heads	[6, 6, 6]
Window size	8
MLP ratio	2.0
Drop path rate	0.1
Total parameters	$\sim 4.7\text{M}$

TABLE III
QUANTITATIVE COMPARISON ON LOL DATASET

Method	PSNR $\uparrow$	SSIM $\uparrow$
<i>Classical Methods (2017-2021)</i>		
SID	13.24	0.442
DeepUPE	13.27	0.452
RF (Retinex)	14.05	0.458
FIDE	16.85	0.678
3DLUT	17.59	0.721
A3DLUT	18.19	0.745
EnlightenGAN (EG)	18.23	0.617
RetinexNet	18.37	0.723
LCDR	18.57	0.641
<i>Recent Transformer Methods (2022-2023)</i>		
LLFormer [13]	23.65	0.820
SNR-Net [11]	24.61	0.842
Retinexformer [12]	25.16	0.855
SwinLLIE (Ours, 4.7M params)	18.66	0.840

C. Quantitative Results

Table III presents the quantitative comparison on the LOL dataset [7]. Baseline results are cited from the CVPR 2023 benchmark [10].

While recent transformer methods achieve higher PSNR, SwinLLIE demonstrates that window-based attention mechanisms can achieve competitive perceptual quality (0.840 SSIM) with significantly fewer parameters than large-scale models (4.7M vs 39.1M for SNR-Net). Compared to classical methods, SwinLLIE achieves +0.29 dB PSNR improvement over RetinexNet and significantly outperforms EnlightenGAN in SSIM (0.840 vs 0.617) [15], demonstrating effective structural preservation.

Table IV provides a parameter efficiency comparison among representative methods.

TABLE IV
MODEL EFFICIENCY COMPARISON

Method	Params	PSNR	SSIM
RetinexNet	0.84M	18.37	0.723
SwinLLIE (Ours)	4.7M	18.66	0.840
SNR-Net	39.1M	24.61	0.842

D. Qualitative Analysis

Fig. 2 demonstrates the visual enhancement results comparing SwinLLIE with baseline methods. The proposed method produces more natural brightness balance and improved texture detail retention, to maintain structural integrity.

E. Discussion

The hybrid loss function combining L1, VGG perceptual, color consistency, edge preservation, and exposure control losses enables the model to balance multiple objectives simultaneously. The L1 loss ensures pixel-level reconstruction accuracy, while the VGG perceptual loss [9] maintains high-level feature similarity. The color consistency loss prevents the

[Qualitative Comparison Figure]

Insert comparison images here showing:

- (a) Input (Low-Light)
- (b) RetinexNet
- (c) EnlightenGAN
- (d) SwinLLIE (Ours)
- (e) Ground Truth

Fig. 2. Qualitative comparison of SwinLLIE against existing methods on the LOL dataset. Proposed method produces more natural brightness balance and improved texture details in comparison to other methods.

common issue of color shifts in enhanced images, and the edge loss preserves structural boundaries that are often degraded in low-light conditions.

Limitations. While SwinLLIE achieves competitive SSIM, the PSNR gap compared to recent methods like Retinexformer suggests room for improvement in pixel-level reconstruction. Additionally, the current evaluation is limited to the LOL dataset; future work should validate on diverse benchmarks such as LOL-v2 and SID.

VI. CONCLUSION

This paper presents SwinLLIE, a novel Swin Transformer-based method for low-light image enhancement. Through leveraging multi-scale features via a U-Net encoder-decoder architecture with RSTB blocks, the method achieves a PSNR of 18.66 dB and SSIM of 0.84 on the LOL dataset while maintaining a compact model size of approximately 4.7M parameters.

The window-based self-attention mechanism provides reduced computational complexity [1], making the model suitable for practical applications. The hybrid loss function effectively balances reconstruction accuracy with perceptual quality, preventing common issues such as color shifts and over-exposure. Future works include further optimizing for edge-device integration, exploring unsupervised training paradigms, and extending the approach to video enhancement.

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